UNIVERSITY OF PERADENIYA
DEPARTMENT OF PHYSICS

END SEMESTER EXAMINATION
SEMESTER II - 2015/2016

PH 102 - General Physics II
PART II

Time allowed: *Two hours*

Answer **Five** questions selecting **Two** from Section **A**, **Two** from Section **B** and **One** from Section **C**.
(Use of calculators are allowed)

Electron charge (e)	$= 1.60 \times 10^{-19}$ C
Rest mass of electron (m_o)	$= 9.11 \times 10^{-31}$ kg
Permittivity of free space (ε_o)	$= 8.85 \times 10^{-12}$ F m^{-1}
Permeability of free space (μ_o)	$= 4\pi \times 10^{-7}$ H m^{-1}
Planck constant (h)	$= 6.63 \times 10^{-34}$ J s
Speed of light in vacuum (c)	$= 3.00 \times 10^{8}$ m s^{-1}
Boltzmann constant (k)	$= 1.38 \times 10^{-23}$ J K^{-1}
Avogadro number (N_A)	$= 6.02 \times 10^{23}$ mol^{-1}.

(Answer Section A in separate books)

SECTION - A

1. (a) Clearly identifying the quantities involved, state the Biot-Savart law in vector form.

(b) Derive an expression for the magnetic flux density at point P at a distance x on the perpendicular bisector of a straight wire of length l carrying steady current I as shown in Figure (1). Note that CD is a finite wire and lengths l and x are in the same order.

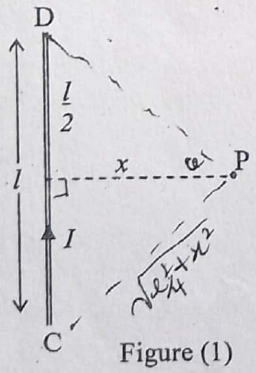


Figure (1)

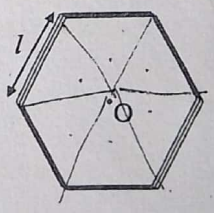


Figure (2)

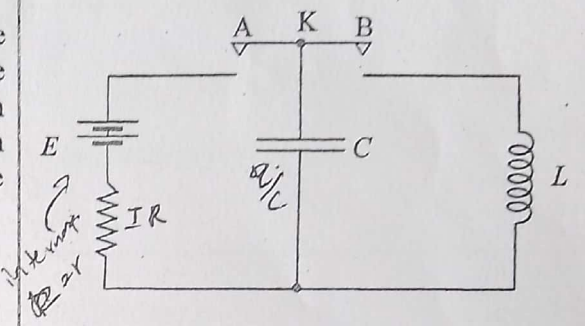
(c) Obtain an expression for the magnetic flux density at the centre O of a hexagonal coil of side length l and N number of turns as shown in Figure (2). If the length $l = 2.0$ cm and $N = 50$ find the magnetic flux density at O.

2. (a) Define mutual inductance and self-inductance.

(b) Derive an expression for the self-inductance per unit length of a coaxial cable with inner conductor of radius a and outer conductor of radius b. Assume that the conductors are very thin and they carry equal currents I in opposite directions.

If $a = 0.5$ mm and $b = 2.0$ mm find the self-inductance per unit length of the cable.

3. The capacitor of capacitance C shown in the circuit is initially charged by pressing end A of the two-way key K at time $t = 0$. The battery has an e.m.f. E and internal resistance r. Obtain an expression for charge q accumulated in the capacitor C at the end of time t.



(i) Sketch the variation of q with t.

(ii) What is the time constant of this part of the circuit?

(iii) Also write down an expression for current I through the circuit at time t.

(iv) If the e.m.f of the cell $E = 3.0$ V and capacitance $C = 2.0$ mF and $r = 50\ \Omega$ find the value of the time constant τ and the current in the circuit at time $t = 2\tau$.

(v) After charging the capacitor fully, the end B of the key is pressed to connect the capacitor in series with the inductor of inductance L. Write down an expression for the resonance frequency of the second circuit. If the value of self-inductance $L = 8$ mH calculate the resonance frequency of the second circuit.

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SECTION B

4. (a) Write down Bohr's postulates.

(b) The quantized electron energy levels of a H-atom is given by

$$E_n = -\frac{me^4}{8\varepsilon_o^2 h^2}\frac{1}{n^2} = -\frac{13.6}{n^2}\ \text{eV}\ ,\ \text{where } n = 1, 2, 3, 4, \ldots\ldots$$

and other symbols have their usual meanings.

(i) What is the ionization energy of H-atom?
(ii) Modify the above equation for a singly-ionized atom with Z protons in the nucleus.
(iii) Find the shortest and the longest wavelengths of the Lyman series of singly-ionized helium.

(c) (i) Briefly explain how the X-rays are produced.
(ii) Give one of the uses of characteristic lines of the X-ray spectra.
(iii) Calculate the minimum X-ray wavelength produced in bremsstrahlung by electrons that have been accelerated through a potential of 10^4 V.

5. (a) (i) State the importance of Compton effect in terms of quantum mechanics.
(ii) Briefly explain what is meant by Compton shift.
(iii) The Compton shift is given by $\lambda' - \lambda = \frac{h}{mc}(1 - \cos\theta)$. Identify the symbols in the equation and calculate the Compton wavelength.
(iv) In a Compton experiment, two types of radiations namely UV and X-rays are used. Which one of them will give the largest fractional shift in wavelength ($\Delta\lambda/\lambda$) at the same scattering angle? Justify your answer. Name the symbols you use.

(b) (i) State the importance of the Davisson-Germer experiment.
(ii) If Davisson and Germer had used a potential of 34 V to accelerate the electrons, would they be able to observe the diffraction pattern? Justify your answer.

(c) The speed of an electron is measured to be 5.00×10^3 m s^{-1} to an accuracy of 0.003%. Find the minimum uncertainty in determining the position of this electron.

6. (a) Find the total binding energy and the binding energy per nucleon, for $^{232}_{90}$Th.

(b) List the conservation laws satisfied in nuclear reactions.

(c) Consider the decay process of ^{232}Th into radium (Ra) by emitting an α particle. The half life of this process is 1.4×10^{10} years.
The atomic mass of ^{232}Th = 232.0381 u, Ra = 228.0311 u and α particle = 4.0026 u.
(i) Write down an equation for the decay process.
(ii) Calculate the Q-value associated with the above decay. Is this an exothermic or endothermic process?
(iii) What is the decay constant of ^{232}Th?
(iv) Suppose we had a 1.00 g sample of ^{232}Th. What is its activity?
(v) How many decays per second occur when the sample is one week old?

SECTION – C

7. (a) The potential energy for a system of two atoms can be represented by an expression of the form $U(r) = \frac{A}{r^3} - \frac{B}{r}$, where A and B are constants.

(i) Find the equilibrium separation, r_o, of the two atoms in terms of A and B

(ii) Determine the binding energy of the system in terms of A and B.

(iii) Determine the corresponding spring constant of this system.

(b) (i) Explain briefly the electrical properties of metals, insulators, and semiconductors using the differences in their band structures.

(ii) Assuming the average distance traveled by electrons between collisions to be roughly the distance between atoms, estimate the electrical conductivity of copper (Cu) at 27 °C. The nearest neighbour spacing between atoms in the Cu lattice is 0.256 nm. Assume that one free electron is released to the lattice by each Cu atom. Atomic mass of Cu is 63.5 g mol^{-1} and density of Cu is 9.0×10^3 kg m^{-3}.

8. (a) (i) HCl molecule is in its second rotational energy level, corresponding to rotational quantum number $J = 2$. If the distance between its nuclei is 0.1275 nm, what is the angular speed of the molecule about its center of mass?
(atomic mass of H = 1.0079 u and Cl = 35.453 u)

(ii) How can the analysis of the rotational spectrum of a molecule lead to an estimate of the size of that molecule?

(b) The effective force constant of a vibrating HCl molecule is $k = 480$ N m^{-1}. A beam of infrared radiation of wavelength 6.203 nm is directed through a gas of HCl molecules. Can the molecules be excited from the ground vibrational state to the first excited vibrational state? Show the mathematical calculations related to your answer.

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